Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/19/2007
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Project Title:	EXPLORATION AND DEVELOPMENT OF A NOVEL MICROPROPULSION TECHNIQUE FOR NAVIGATION IN LOW GRAVITY ENVIRONMENTS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	20373-001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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Flight Program:			
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COI Name (Institution):	Gamezo, Vadim (Naval Research Laboratory)		
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	This work is based on a micropropulsion idea generated from a relatively recent discovery concerning the behavior of laminar flames propagating in very narrow tubes. When a slow laminar flame is ignited near the closed end of a tube, the material ahead of the flame is accelerated and a boundary layer is formed along the walls in the unreacted material. Due to the presence of the boundary layer, the flow velocity changes across the channel, increasing from zero at the wall to a maximum in the middle of the tube. Previous simulations have shown that the nonuniform flow stretches the flame, so that the shape of the flame becomes similar to the velocity profile. This increases the flame surface area, thus accelerating the energy generation and the flow. There is little change in the laminar burning rate, but the speed of the flame that propagates with the flow grows very rapidly in the laboratory frame of reference, from centimeters per second to hundreds of meters per second for adiabatic walls.		

Task Description:	The high-speed flow created by the accelerating flame has the potential to provide a significant thrust that can be further enhanced by attaching an appropriately shaped exist nozzle. The thrust will gradually increase as the flame and flow accelerate, and reach a maximum when the flame approaches the open end of the tube or the nozzle. We consider regimes in which the flame remains laminar until the end of the tube and does not produce strong shocks or detonations. In contrast to the majority of other combustion-based propulsion concepts, most of the thrust here can be provided by the material accelerated ahead of the flame. Therefore, a propulsion device could be only partially filled with a combustible mixture. The rest of the material can be an inert gas or combustion products from a previous pulse. Task Objective: To explore the possibility of using flame acceleration in narrow tubes for micropropulsion. This micropropulsion concept is of particular interest for low-gravity environments, where relatively small, controlled thrusts could be used for navigational corrections. In such cases, there might not be a need for the high repetition rates required for pulse combustion and pulse detonation engines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The micropropulsion concept has a potential to provide significant benefits for space exploration and satellite navigation.
Task Progress:	A flame propagating towards the open end of a partially obstructed channel filled with a gaseous combustible mixture can accelerate or oscillate, depending on the channel width and obstacle configurations. The accelerating flame is able to produce a high-speed flow at the open end of the channel that has the potential to provide significant thrust, and this can be used in micropropulsion devices. We study these phenomena for channels with obstacles using multidimensional reactive Navier-Stokes numerical simulations. Computations show that at initial stages, the flame and flow acceleration is caused by thermal expansion of hot combustion products. At later stages, shock-flame interactions, Rayleigh-Taylor, Richtmyer-Meshkov, and Kelvin-Helmholtz instabilities, and flame-vortex interactions in obstacle wakes become responsible for the increase of the flame surface area, the energy-release rate, and, eventually, the shock strength. Computations performed for different channel widths d with the distance between obstacles d and the constant blockage ratio 0.5 reproduce the main supersonic regimes observed in experiments: choking flames, quasi-detonations, and detonations. For quasi-detonations, both the initial DDT and succeeding detonation reignitions occur when the Mach stem, created by the reflection of the leading shock from the bottom wall, collides with an obstacle. As the size of the system increases, the time to DDT and the distance to DDT increase linearly with d2. We also observe an intermediate regime of fast flame propagation in which local detonations periodically appear behind the leading shock, but do not reach it. Channels with obstacles are of a particular interest for micropropulsion because they can provide simple ways to control combustion regimes and trust development by changing the channel width, the obstacle spacing, or the fuel/oxydizer ratio.
Bibliography Type:	Description: (Last Updated: 07/20/2007)
Abstracts for Journals and Proceedings	Gamezo VN, Ogawa T, Oran ES. "Numerical Simulations of Flame Acceleration and Transition to Detonation in Channels with Obstacles." 11th International Conference on Numerical Combustion, Granada, Spain, April 23-26, 2006. Abstracts, 11th International Conference on Numerical Combustion, 2006. , Apr-2006
Articles in Peer-reviewed Journals	Gamezo VN, Oran ES. "Flame acceleration in narrow channels: applications for micropropulsion in low-gravity environments." AIAA Journal, 2006 Feb; 44(2):329-36. , Feb-2006
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Papers from Meeting Proceedings	Gamezo VN, Ogawa T, Oran ES. "Deflagration-to-detonation transition in premixed H2-air in channels with obstacles." 45th AIAA Aerospace Sciences Meeting & Exhibit, Reno, NV, January 8-11, 2007. 45th AIAA Aerospace Sciences Meeting & Exhibit, Paper AIAA-2007-1172, 2007. , Jan-2007
Papers from Meeting Proceedings	Gamezo VN, Oran ES. "Flame Acceleration in Narrow Tubes: Applications for Micropropulsion in Low-Gravity Environments." 43rd AIAA Aerospace Sciences Meeting & Exhibit, Reno, NV, January 10-13, 2005. 43rd AIAA Aerospace Sciences Meeting & Exhibit, Paper 2005-0540, 2005. , Jan-2005